Intermediate elective

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Set up

Packages

```r
# general use and cleaning
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(here)
```

```
here() starts at /Users/audreyjuneman/Desktop/UCSB/ENVS/ENVS-193DD/github/intermediate-
elective
```

```r
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```
library(snakecase)

# wrapping axis labels
library(scales)
```

```
Attaching package: 'scales'

The following object is masked from 'package:purrr':

    discard

The following object is masked from 'package:readr':

    col_factor
```

```
# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)

#colors
library(NatParksPalettes)
```

Data

```
veg <- read_csv(here("data", "veg.csv"))
```

```
Rows: 27036 Columns: 17
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (10): monitors, transect_name, vp_axis, site, transect_side, vp_zone, q...
dbl   (6): x1, transect_length, num_quad, transect_distance, percent_cover, ...
date  (1): date

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

```
Rows: 257 Columns: 4
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (2): year_pool, transect_name
dbl (2): year, num_quad

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# create a new object from veg
pools_implicitNA <- veg |>
  # filter all occurrences where transect_name contains VP
  filter(str_detect(transect_name, "VP")) |>
  # group by year, pool, and species
  group_by(year, transect_name, psoc) |>
  # calculate mean percent cover
  summarize(mean_pc = mean(percent_cover, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called year_pool from year and transect name
  unite("year_pool", year, transect_name, remove = TRUE) |>
  # join with metadata by year_pool
  left_join(metadata, by = "year_pool")
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by year, transect_name, and psoc.
i Output is grouped by year and transect_name.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(year, transect_name, psoc))` for per-operation
  grouping (`?dplyr::dplyr_by`) instead.
```

```r
# create a new object from veg
pools_explicit0 <- veg |>
  # filter all occurrences where transect_name contains VP
  filter(str_detect(transect_name, "VP")) |>
  # complete all possible combinations of year, pool, psoc and distance
  complete(year, transect_name, psoc, transect_distance,
           # fill values of percent cover with 0
           fill = list(percent_cover = 0)
           ) |>
  # group by year, pool, and psoc
  group_by(year, transect_name, psoc) |>
  # calculate total percent cover across all quads
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
  # ungroup the data frame
  ungroup() |>
  # create a new column called year_pool from year and transect name
  unite("year_pool", year, transect_name, remove = TRUE) |>
  # join with metadata data frame
  left_join(metadata, by = "year_pool") |>
  # calculate mean percent cover from total percent cover/number of quadrats
  mutate(mean_pc = sum_pc/num_quad)
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by year, transect_name, and psoc.
i Output is grouped by year and transect_name.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(year, transect_name, psoc))` for per-operation
  grouping (`?dplyr::dplyr_by`) instead.
```

```r
# creating new ojbect from implicit NA data frame
filtered_implicitNA <- pools_implicitNA |>
  # filter to only include
  filter(psoc == "Eryngium_vaseyi")

filtered_explicit0 <- pools_explicit0 |>
  filter(psoc == "Eryngium_vaseyi")

# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```
Rows: 50 Columns: 15
```

```
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (15): verbatim_name, taxonRank, scientificName, kingdom, Phylum, Class, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data",
                                "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
```

```
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter
  # (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
```

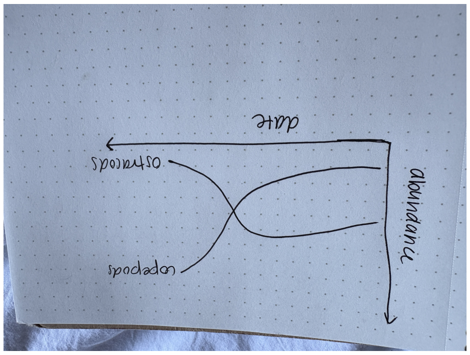
```r
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal,
                                 .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by date, site, and taxon.
i Output is grouped by date and site.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(date, site, taxon))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

#1. Explain your inspiration

- For my Intermediate elective, I chose the coal falls and renewables rise visualization, due to my interest in energy usage in the United States. I enjoy the color scheme and how easy to interpret the graph is due to the nice colors and organization.
- I think this makes sense for comparing ostracods and copepod abundance throughout time, because it compares two specific variables and shows their trends throughout time. I will specificlaly be looking at Devereaux creek.
- My x axis will be `date_site` and my y-axis will be `taxon`. It will be colored by `taxon`.

#2. Plan your figure



#3. Code your figure

```
#filtering dataset for copepod line
aq_ins_dev <- aq_ins_clean |>
  filter(site_full == "Devereux Creek")
```

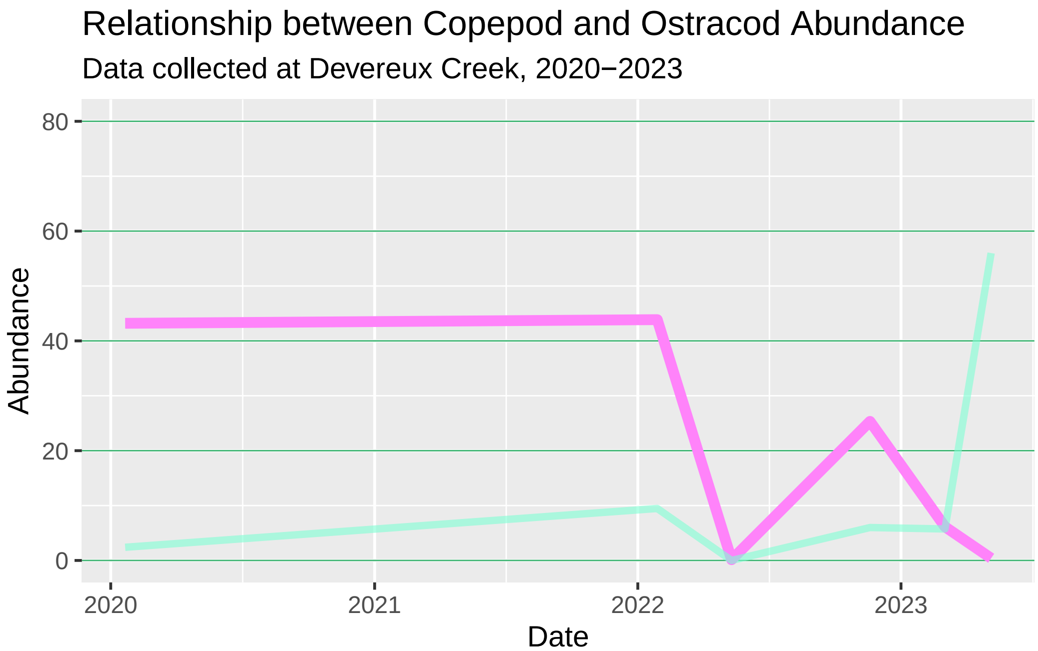
```
#| label: plot
#| warning: false

### |- main plot ----
p <- aq_ins_dev |>
  ggplot(aes(x = date, y = ave_lit, color = taxon)) +

  #creating y intercept lines
  geom_hline(
    yintercept = seq(0, 80, by = 20),
    color = "mediumseagreen",
    linewidth = 0.25
  ) +
  # creating copepod line
  geom_line(
    data = aq_ins_dev |> filter(taxon == "copepod"),
    linewidth = 1.9,
    color = "orchid1",
    alpha = 1.0
  ) +
  #creating ostracod line
  geom_line(
    data = aq_ins_dev |> filter(taxon == "ostracod"),
    linewidth = 1.3,
    color = "aquamarine",
    alpha = 0.60
```

```r
  ) +
  geom_point(
    data = aq_ins_dev |> filter(date == 2020-01-21),
    size = 3.5,
    shape = 21,
    fill = "pink",
    stroke = 1.8
  ) +
  geom_point(
    data = aq_ins_dev |> filter(date == 2020-01-21),
    size = 3.5,
    shape = 21,
    fill = "pink",
    stroke = 1.8
  ) +
  # Labs
  labs(
    title = "Relationship between Copepod and Ostracod Abundance",
    subtitle = "Data collected at Devereux Creek, 2020-2023",
    x = "Date",
    y = "Abundance"
  )

p
```



Relationship between Copepod and Ostracod Abundance
Data collected at Devereux Creek, 2020−2023
Abundance
80
60
40
20
0
2020
2021
2022
2023
Date